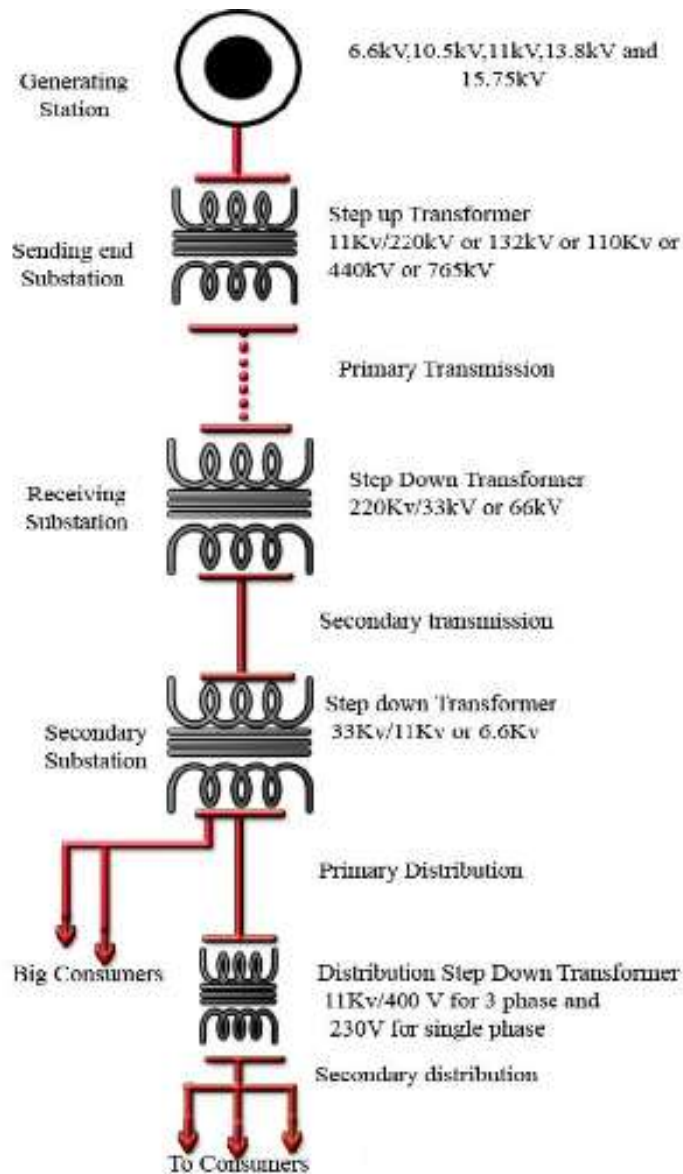


Module 4



- **Generating station** : Electric power is produced in generating station by 3-phase alternators operating in parallel. The usual generation voltage is 11 kV. This 11 kV is stepped up to 110kv or 132kV or 220kV or 440kV at the generating station with the help of 3-phase power transformers.
- **Primary Transmission** : The electric power at 110kv or 132kV or 220kV or 440kV or 765kV is transmitted by 3-phase, 3-wire overhead system to the outskirts of the city. This forms the primary transmission
- **Secondary Transmission** : The primary transmission line terminates at the receiving station (RS) where the voltage is reduced to 33kV or 66kV by step-down transformers. From receiving station, electric power is transmitted to various sub-stations (SS) and this forms secondary transmission
- **Primary distribution**: At sub-station (SS) voltage is reduced to 11kV using step-down transformers. The 11 kV lines run along the important road sides of the city. It may be noted that big consumers (having demand more than 50 kW) are generally supplied power at 11 kV for further handling with their own sub-stations
- **Secondary distribution** : Primary distribution line (11 kV) terminates at distribution sub-stations (DS) or distribution transformer. These sub-stations are located near the consumers' localities and step down 11kV to 400V which is distributed to the consumers using 3-phase, 4-wire system (440 V between phases and 230 V between any phase and neutral)

Power stations equipment

1. Alternators
2. Exciters
3. Power Transformers
4. Bus bar
5. Voltage regulators
6. Switchgears
7. Protective devices
8. Current & Potential transformers
9. Control panel
10. Batteries

Alternators:

Connected in parallel with same terminal voltage.

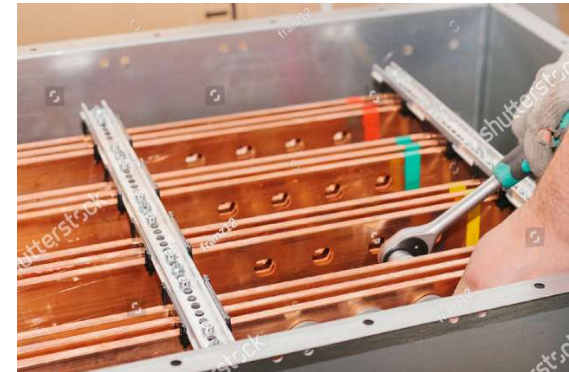
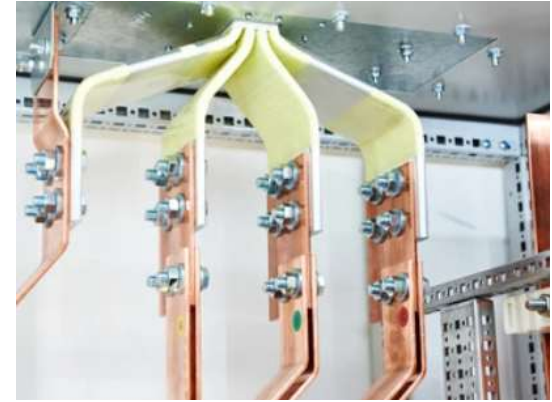
Exciters :

- The excitation system is required to provide necessary field current to the rotor winding of the synchronous machine
- Loss of excitation unit results in
 - Remaining units to pick up the drop
 - Supply large reactive current drawn by the unexcited unit.
- Amount of excitation required depends on
 - Load current, load power factor , speed of the machine
- Excitation system can be - Individual or centralized

Bus bars

Main bar or conductor carrying an electric current to which many connections are made

- They are usually bare rectangular x-section bars but can be of other shapes as round tubes, round solid or square tube bars
- They are usually Aluminum
 - Higher conductivity on weight basis
 - Lower cost for equal current carrying capacity
 - Excellent resistance
 - Ease of formability
- The most common sizes are
 - 40 x 4mm, 40 x5mm, 50x 5mm,50x6mm, 60x 8mm,80x8mm, 100x10mm, according to the current to be carried
- In some cases two buses are provided to which incoming & outgoing feeders and the principal equipment may be connected.
 - Main bus – elaborate system of measuring instruments relays associated with it
 - Auxiliary bus



Voltage Regulators

- Automatic voltage regulators work by stabilizing the output voltage of generators at variable loads, but can also divide the reactive load between generators that are running in parallel , and helps the generator respond to overloads.
- Functions are:
 - Control of system voltage within prescribed limit
 - Proper division of reactive power between the alternators operating in parallel
 - Prevention of dangerous over voltage on the occurrence of sudden loss of load on the system
 - Increase of excitation under system fault condition, to prevent the loss of synchronism

Power Transformers

- Power transformers are used for stepping up the voltage for transmission at generating station.

Switch gears

- During operation of power system the generating plants transmission lines distributors and other electrical equipment are required to be switched on or off under both normal and abnormal operating conditions.
- The apparatus including its associated auxiliaries employed for controlling,regulating or switching on or off the electrical circuits in the electrical power system is called switch gears

Protective equipment

- They are installed to function under abnormal conditions to prevent failure or isolate trouble and limits its effect
- It must function reliably and quickly
- Fuses, circuit breakers and relays, lightning arresters, insulators etc

Current transformers and Potential transformers

- Used in every circuit for measuring the current and voltage values respectively

Control room and control panels

- Act as the nerve centre for power stations
- Performs operations like voltage adjustment, load control, emergency tripping of turbines etc
- Main equipment in control room are synchronizing equipment, voltage regulator, relays, ammeter, voltmeter, wattmeter, KWh meter, KVARh meter, temperature gauges, water level indicator etc.

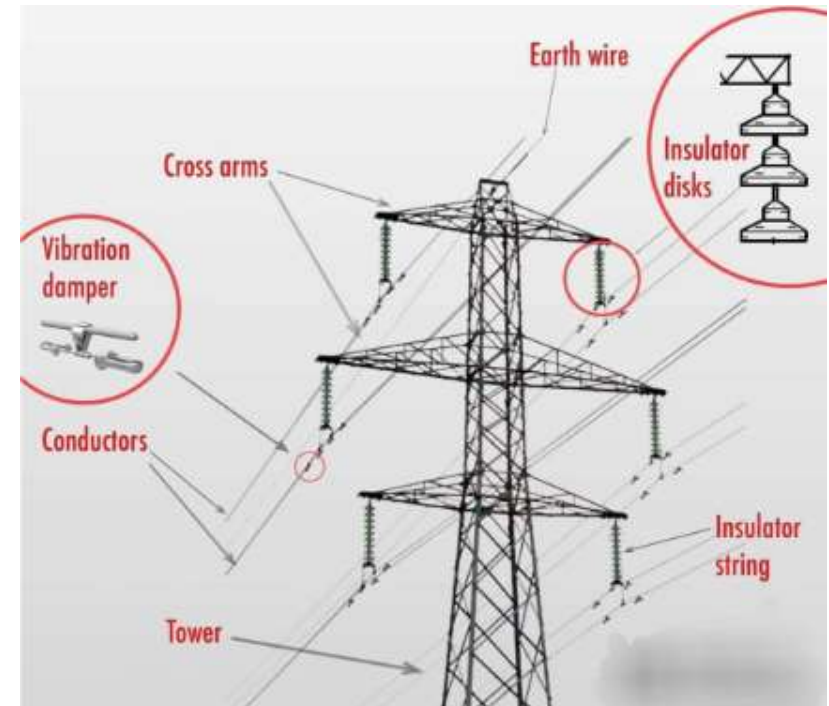
Batteries

- All power plants and substations require dc supply for protection and control purpose and DC supply is obtained from storage batteries
- Two types : lead acid and alkaline batteries
- Lead acid batteries are commonly used in power stations because of higher cell voltage and lower cost

Overhead transmission line

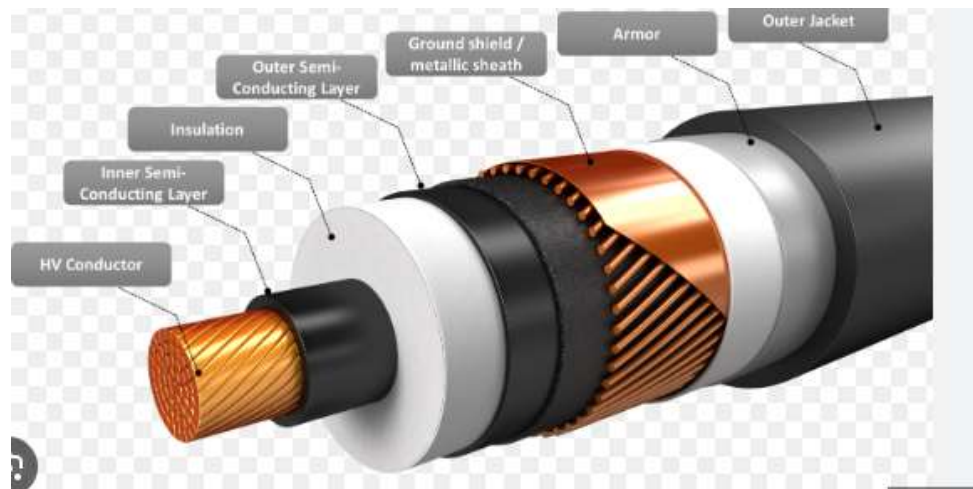
- Overhead transmission line uses bare conductors.
- These conductors placed at a height from the ground. To maintain clearance between the conductors and ground, supporting towers are used.
- The voltage of the transmission line decides the height of the tower.
- The insulators used to provide insulation between the conductor and the tower.

Operating voltage in kV	Number of discs in	
	Suspension assembly	Dead end assembly
11	1	1
33	2	3
66	5	6
132	9	10
220	14	15
400	21	22



Underground transmission line

- In the underground transmission system, the number of conductors bunched together with proper insulation.
- The underground cable provided with lead sheet and armoring. These provide protection against moisture and mechanical injury.
- As the voltage level increase, the thickness of insulation increases.



Substation and classification of substation

Main function of substation are

- (i) To receive energy transmitted at high voltage from generating station
- (ii) To reduce the voltage to a value appropriate for local distribution
- (iii) To provide facilities for switching.
- (iv) They provide points where safety devices may be installed to disconnect equipment or circuit in the event of fault.



Classification

1. According to service requirement

(a) **Transformer substation:-** changes voltage level

(b) **Power factor correction substation:-**
Improves powerfactor

(c) **Converting substation**

Changes AC to DC power

2. According to constructional features

(a) **Indoor substation:-** substation equipments are installed indoor upto 11kV

(b) **Outdoor substation:-** For voltages beyond 66kV, equipments are installed outdoor.

(c) **Underground substation:-** equipments are placed underground. In thickly populated areas and price of land is high.

SUBSTATION EQUIPMENTS

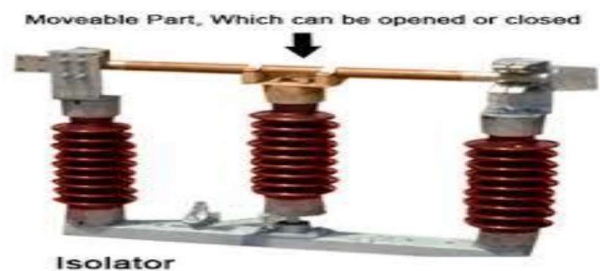
1. Power transformers

Power transformers are used in transmission network of higher voltages for step- up and step down application (400 kV, 200 kV, 110 kV, 66 kV, 33kV) and are generally rated above 200MVA



2. Isolator

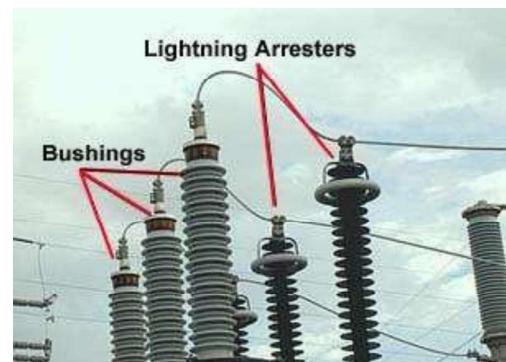
- It is a disconnection switch and to be operated on no load manually.
- It is used to make sure that an electrical circuit can be completely de-energized for service or maintenance.
- It provide electrical isolation of the equipment .



3. Lightning Arrester

➤ It discharges the over voltage surges to earth and protects the equipment insulation from switching surges and lightning surges.

➤ It is located at the starting of the substation as seen from incoming transmission lines and is the first equipment of the substation



4. Instrument transformers

The main purpose of an instrument transformer is to decrease high current as well as voltages for a secure & realistic value. These values can be calculated with conventional devices. This transformer is also used for triggering the protective relay (AC type) by providing the current as well as voltage. They are of

a. Potential transformer

Potential transformers are instrument transformers. They have a large number of primary turns and a few number of secondary turns.

It converts voltages from high to low so that meters can handle

a. Current transformer

- CT is a type of instrument transformer that is used in power system for measurement, detection, protection of the system.
- Current transformers are used extensively for measuring current and monitoring the operation of the power grid.



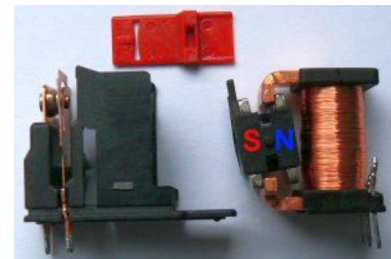
5. Circuit breaker

- It designed to protect an electrical circuit from damage caused by overload or short circuit.
- Its basic function is to detect a fault condition and discontinue the electric flow
- It includes two moving parts that are usually closed. When an error happens in the system, then the relay transmits the signal to the circuit-breaker & therefore their parts are moved separately.



6. Relays

The relay is an electrical device, and the main role of this device in the substation is, it guards the grid component against the irregular conditions like faults. This is one type of detecting device, used to detect and determine the fault location, and then it sends the signal to the circuit breaker. After receiving the signal from the relay, the circuit breaker will detach the faulted part.

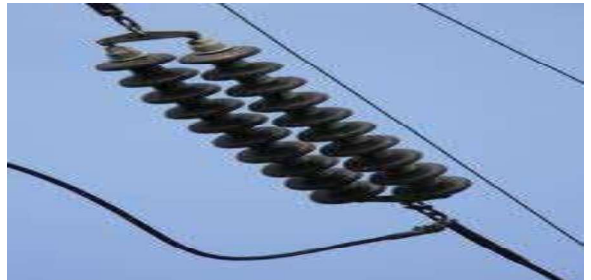


6. Busbar

The bus bar is a very important component in an electrical substation. It is a kind of current carrying conductor where many connections are made

7. Insulators

- An insulator, also called a dielectric, is a material that resists the flow of electric current.
- Insulators are used in electrical equipment to support and separate electrical conductors without allowing current through themselves



8. Batteries

In large power stations or substations, the operation of lighting, relay system, or control circuits are powered by batteries.

9. Capacitor bank

This device is inbuilt with capacitors that are connected either in series or else parallel. The main function of this is to store the electrical energy in electrical charge form. The capacitor bank is an efficient method to improve power factor



10 .Wavetrap

❑ It trap the high frequency communication signals sent on the line from the remote substation and diverting them to the telecom/teleprotection panel in the substation control room .

❑ The Line trap offers high impedance to the high frequency communication signals thus obstructs the flow of these signals in to the substation bus bars.

❑ The signals are primarily teleprotection signals and in addition, voice and data communication signals.



Metering, controlroom and relay panels

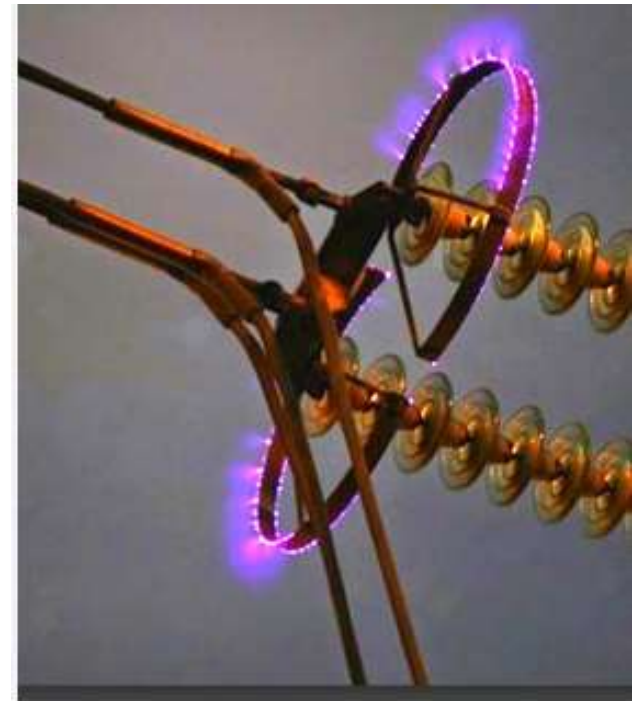


- To house various measuring Instruments, control Instruments, Protective relays. They are located in air-conditioned building. Control Cables are laid between Switchyard equipment and these panels.

Corona and skin effect

Corona

- When an alternating potential difference is applied across two conductors, whose spacing is large in comparison with their diameter, then the atmospheric air surrounding the conductor is subjected to electrostatic stresses.
- For an increase in potential difference, a stage arrives when a faint luminous glow of violet colour appears together with a hissing noise, accompanied with the production of ozone. This phenomenon is called corona.
- This is very much evident in transmission line above 100kV
- In EHV AC transmission line the visible glow of corona is mostly uniform about both conductors but for HVDC transmission line glow is uniform and brighter about the positive conductor but spotty about negative conductor



Effect of Corona

- A violet glow is observed
- Produces a hissing sound
- Produces ozone readily detected by its characteristic odour
- Glow is maximum over rough and dirty surfaces of the conductor
- Accompanied by power loss
- Charging current under corona condition increases because the corona induces harmonic currents

Factors affecting Corona

1. Atmosphere - Affected by the physical state of the atmosphere
2. Conductor
 - Size (diameter) – corona decreases with increase in conductor diameter
 - Shape (stranded or smooth) – stranded conductor give rise to more corona
 - surface condition (dirty or clean)
3. Spacing between the conductors – with increase in spacing the electrostatic stress get reduced
4. Line voltage – increase in line voltage increase the electrostatic stress between the conductors

Methods to reduce the Corona

1. Increasing the spacing between the conductors
2. Increasing the conductor diameter
 - Using steel cored Aluminium conductor
3. Using Bundled conductors instead of stranded one



4. Using Corona rings

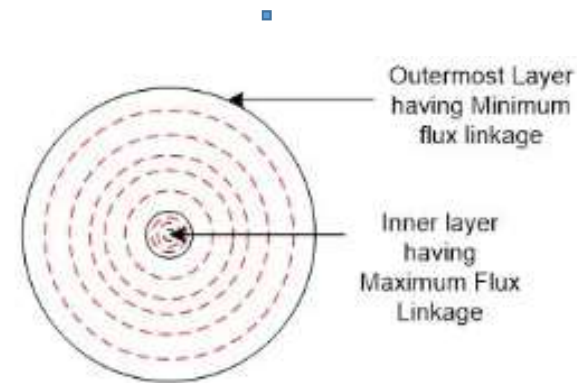


Corona discharge occurs first at the points where the curvature is greatest - i.e. at suspension points, corners and edges. Corona rings are installed at these points to prevent corona formation.

A corona ring is electrically connected to the conductor, encircling the points where corona discharge may occur. Therefore, the corona ring distributes the electric field (or charge) across a wider area, because of its smooth round shape. Hence, it reduces the potential gradient.

Skin effect

- Skin effect is defined as the tendency of an AC current to distribute unevenly over the cross-section of a conductor, such that the current density is highest near the surface of the skin of the conductor and decreases exponentially towards the core.
- This means that the inner part of the conductor carries less current than the outer part, resulting in increased effective resistance of the conductor.
- Depends on
 1. **Frequency** - frequency is directly proportional to the skin effect. Because, Reactance is directly proportional the supply frequency ($X_L = 2\pi f L$)
 2. **Diameter of conductor** - skin effect is very less for thin conductors and very high for thick conductors.
 3. **Shape of conductors** - Skin effect is more in case of Solid conductors is less in case of stranded conductors and Hollow Conductors
 4. **Nature of material** - Relative permeability (μ_r) is a property of material which supports the formation of magnetic field and skin effect is directly proportional to the Relative permeability (μ_r).





Disadvantage of skin effect

- It reduces the effective area of the conductor from which the current is flowing.
- The overall resistance of the conductor is increased.
- Copper loss increases
- Voltage regulation decreases
- Efficiency decreases

Methods to reduce skin effect

- The skin effect is reduced when the conductor of the line has low magnetic permeability.
- Using less size of conductor results in a reduction of skin effect.
- By Using bundled conductor, stranded conductors